GEOGRAPHY

PHYSICAL GEOGRAPHY

GEOMORPHOLOGY

MODULE - 36

DESERT TOPOGRAPHY

- EROSIONAL LANDFORMS
- DEPOSITIONAL LANDFORMS

GEOMORPHOLOGY

PLAN OF ACTION

WHAT GVR SIR HAS COVERED

- EARTH'S INTERIOR
- SOURCES OF HEAT IN EARTH'S INTERIOR
- CONTINENTAL DRIFT THEORY
- PLATE TECTONICS THEORY
- CONVERGENT, DIVERGENT & TRANSFORM PLATE BOUNDARIES

WHAT I WILL COVER

- VOLCANOES
- EARTHQUAKES
- GEOMORPHIC PROCESSES
- LANDFORMS & THEIR EVOLUTION
  1. FLUVIAL LANDFORMS
  2. GLACIAL LANDFORMS
  3. ARID LANDFORMS
  4. KARST TOPOGRAPHY
  5. COASTAL LANDFORMS

THE MECHANISM OF ARID EROSION

1. Arid landforms are the result of many combined factors, one reacting upon the other.
2. Insufficient rainfall (often less than 5 inches) coming at most irregular periods, coupled with very high temperatures (87^0 F is the average) and a rapid rate of evaporation, are the chief causes of aridity.
3. Sub-aerial denudation through the processes of weathering (mechanical and chemical), wind action and the work of water have combined to produce a desert landscape that is varied and distinctive.

THE MECHANISM OF ARID EROSION

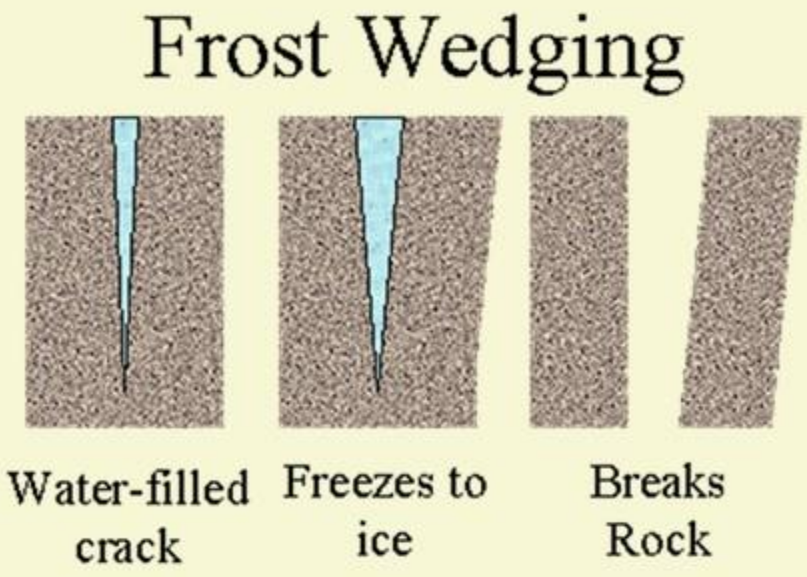
WEATHERING

EXFOLIATION

1. This is the most potent factor in reducing rocks to sand in the arid regions.
2. The heating of the rocks causes the outer surface to expand and so prise itself off from the interior rocks, so that it peels off in successive very thin layers
3. Such an onion-peeling process of mechanical weathering is called exfoliation.

THE MECHANISM OF ARID EROSION

WEATHERING



4 Similarly, when water gets into cracks and joints of rocks and the temperature at night suddenly drops to below freezing point, the water freezes and therefore expands by 10% of its volume.

5 Successive freezing will prise off fragments of rock which accumulate as screes. These rocks fragments become the "teeth" or tools of wind erosion.

ARID TOPOGRAPHY

ACTION OF WINDS IN DESERTS

1. The wind though not the most effective agent of erosion, transportation and deposition, is more efficient in arid than in humid regions.

2. Since there is little vegetation or moisture to bind then loose surface materials, the effects of wind erosion are almost unrestrained.

3. Wind erosion is carried out in the following ways.

ARID TOPOGRAPHY

ACTION OF WINDS IN DESERTS

1 DEFLATION

1. This involves the lifting and blowing away of loose materials from the ground.

QATTARA DEPRESSION

2. Such unconsolidated sands and pebbles may be carried in the air or rolled along the ground depending on the grain size.
3. The finer dust & sands may be removed miles away from their place of origin, and be deposited even outside the desert margins.
4. Deflation results in the lowering of the land surface to form large depressions called deflation hollows.
5. The Qattara Depression of the Sahara Desert lies almost 450 feet below sea level.

ARID TOPOGRAPHY

ACTION OF WINDS IN DESERTS

2 ABRASION

1. The sand-blasting of rock surfaces by winds when they hurl sand particles against them is called abrasion.

2. The impact of such blasting results in rock surfaces being scratched, polished and worn away.
3. Abrasion is most effective at or near the base of rocks, where the amount of material the wind is able to carry is greatest.
4. This explains why telephone poles in the deserts are protected by a covering of metal for a foot or two above the ground.
5. A Great variety of desert feature are produced by abrasion.

ARID TOPOGRAPHY

ACTION OF WINDS IN DESERTS

3 ATTRITION

1 When the wind-borne particles roll against one another in collision they wear each other away so that their sizes are greatly reduced and grains are rounded into millet-seed sand. This process is called attrition.

In the combined processes of abrasion, deflation and attrition, a wealth of characteristic desert landforms emerge.

ARID TOPOGRAPHY

EROSIONAL LANDFORMS

1 ROCK PEDESTALS OR MUSHROOM ROCKS

1 The sand-blasting effect of winds against any projecting rock masses wears back the softer layers so that an irregular edge is formed on the alternate bands of hard and soft rocks.

ARID TOPOGRAPHY

EROSIONAL LANDFORMS

1 ROCK PEDESTALS OR MUSHROOM ROCKS

2. Grooves and hallows are cut in the rock surfaces, carving them into fantastic and grotesque looking pillars called rock pedestals.

3. Such rock pillars will be further eroded near their bases where the friction is greatest.

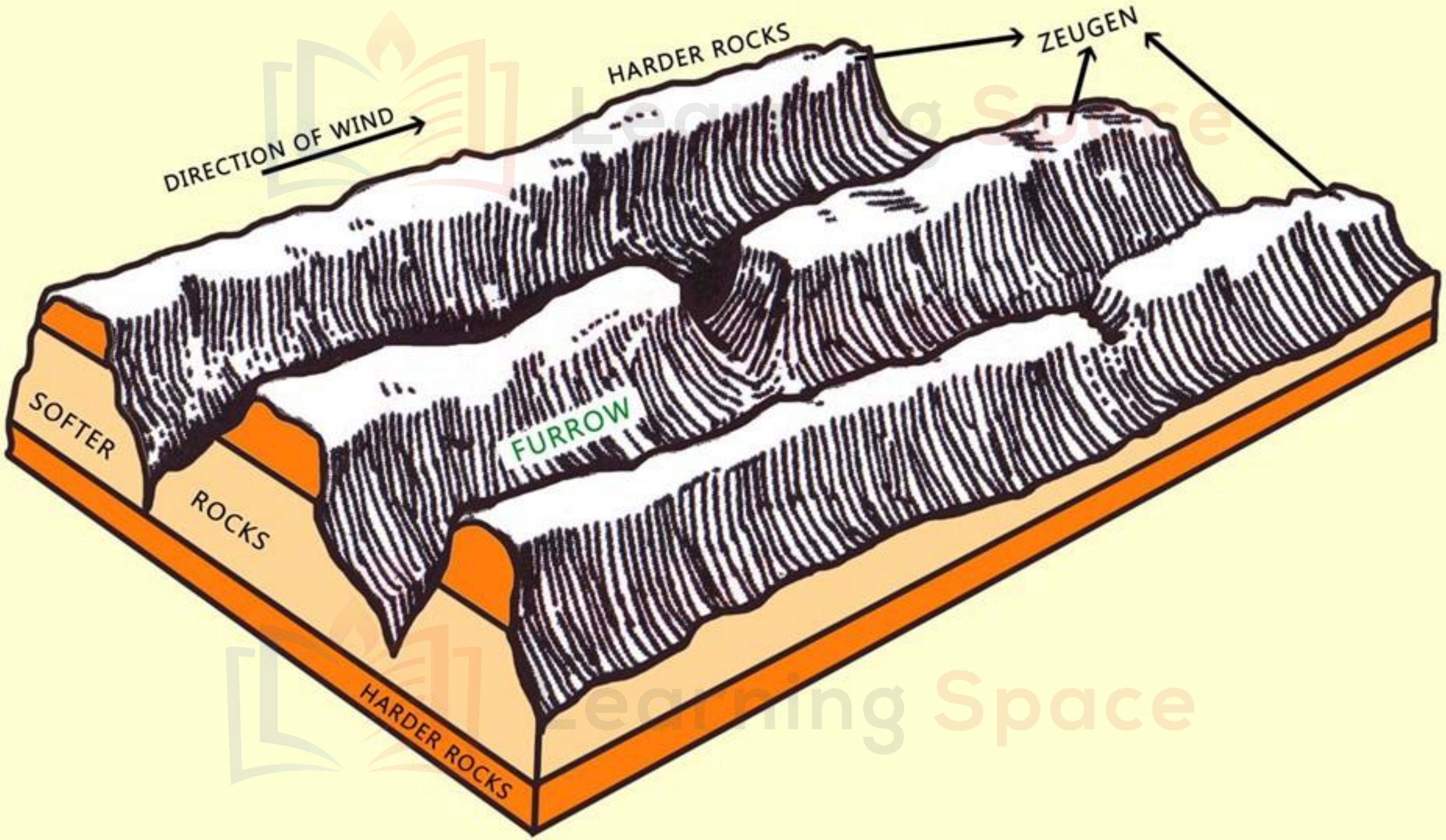
4. This process of undercutting produces rocks of mushroom shape called mushroom rocks.

ARID TOPOGRAPHY

EROSIONAL LANDFORMS

2 ZEUGEN

1. These are tabular masses which have a layer of soft rocks lying beneath a surface layer of more resistant rocks.
2. The sculpting effects of wind abrasion wear them into a weird-looking 'ridge and furrow' landscape.
3. Mechanical weathering initiates their formation by opening up joints of the surface rocks. Wind abrasion further "eats" into the underlying softer layer so that deep furrows are developed. The hard rocks then stand above the furrows as ridges or Zeugen



ZEUGEN WITH HORIZONTAL STRATA OF HARD AND SOFT ROCKS

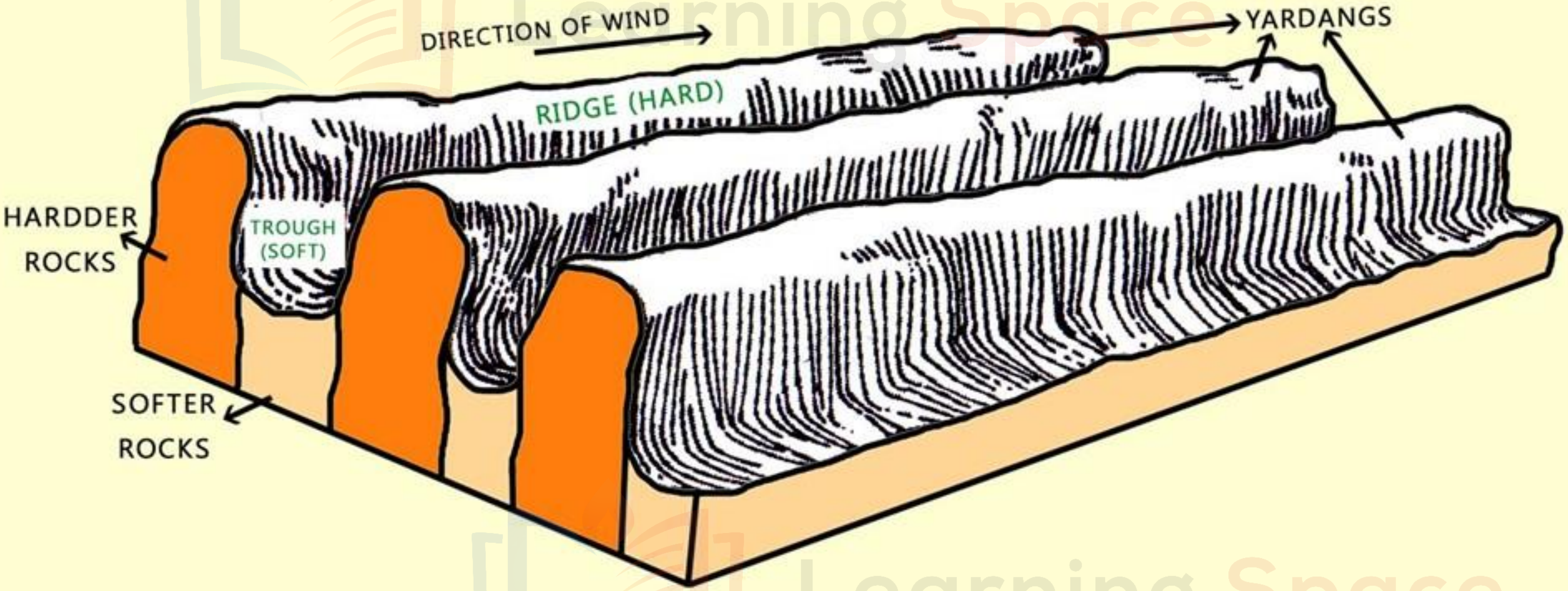
ARID TOPOGRAPHY

EROSIONAL LANDFORMS

3 YARDANGS

1. Quite similar to the “ridge and furrow” landscape of Zeugen are the steep-sided yardangs.

2. Instead of lying in horizontal strata upon one another, the hard and soft rocks of yardangs are vertical bands and are aligned in the direction of the prevailing winds.



YARDANGS WITH VERTICAL BANDS OF HARD AND SOFT ROCKS

ARID TOPOGRAPHY

EROSIONAL LANDFORMS

3 YARDANGS

3 Wind abrasion excavates the bands of softer rocks into long, narrow corridors, separating the steep-sided over-hanging ridges of hard rocks, called Yardangs.

4 They are commonly found in the Atacama Desert, Chile, but the more spectacular one with yardangs rising to 25 to 50 feet are best developed in the interior deserts of Central Asia where the name originated.

ARID TOPOGRAPHY

EROSIONAL LANDFORMS

4 MESAS AND BUTTES

1. Mesa is a Spanish word meaning 'table'. It is a very resistant horizontal top layer, and very steep sides.

2. The hard stratum on the surface resists denudation by both wind and water, and thus protects the underlying layers of rocks from being eroded away.

2. Mesas may be formed in canyon regions eg. Arizona.

ARID TOPOGRAPHY

EROSIONAL LANDFORMS

4 MESAS AND BUTTES

4 Continued denudation through the ages may reduce mesas in area so that they become isolated flat-topped hills called buttes.

5 Many of them in arid countries are separated by deep gorges or canyons.

ARID TOPOGRAPHY

EROSIONAL LANDFORMS

5 INSELBERG

1 This is a German word meaning 'island-mountain'. They are isolated residual hills rising abruptly from the level ground.

2 They are characterized by their very steep slopes and rather rounded tops.

3 They are often composed of granite or gneiss, and are probably the relics of an original plateau which has been almost entirely eroded away.

4 Inselbergs are typical of many desert and semi-arid landscapes of old age e.g. those of northern Nigeria, western Australia and the Kalahari Desert.

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